

Spring-2026

Saturday, 14th March

CSE-345 Computer Architecture and Assembly Language Midterm Exam

Total questions/marks: 40/40

Duration: 120 minutes

Name: _____ Student ERP: _____

Q1. What is the sum of your erp digits? **1M**

Q2. In Figure 1, if the branch instruction at memory location 0x20 is taken, then what instructions will be fetched from the instruction memory in the first 4 clock cycles? **1M**

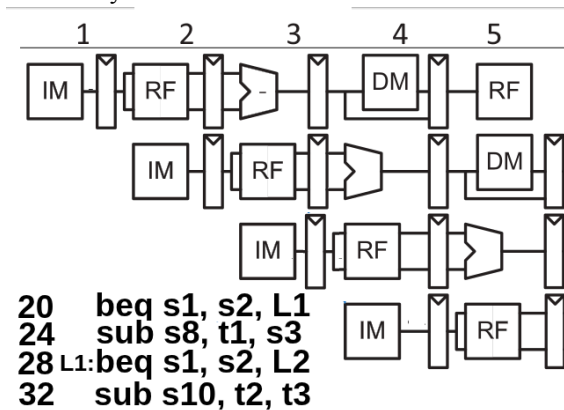


Figure 1

1. beq, sub, beq, sub
2. beq, sub, sub, sub
3. beq, sub, beq, beq
4. None of the rest

Q3. In Figure 1, if both the branch instructions are NOT taken, then what instructions will be fetched from the instruction memory in the first 4 clock cycles? **1M**

1. beq, sub, beq, sub
2. beq, sub, sub, beq
3. beq, sub, beq, beq
4. none of the rest

Q4. The role of the Extend Unit in the RISC-V architecture is to **1M**

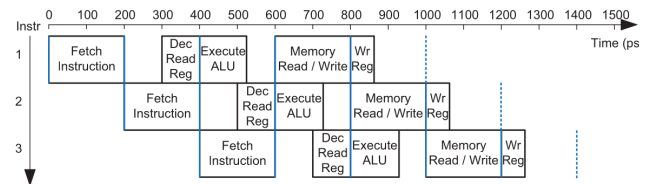
1. Multiply the input bit pattern by 2
2. Convert the input bit pattern into a 32-bit output bit pattern
3. Neither option 1 nor option 2
4. Both option 1 and option 2

Q4. Single instruction multiple data processing paradigm is an example of **1M**

1. Instruction-level parallelism
2. Data level parallelism

3. Thread-level parallelism
4. Memory level parallelism

Q5. Which of the stages will fail if $Fclk = 10GHz$? Ignore the t_{pcq} and t_{setup} **1M**



1. Fetch Only
2. Execute Only
3. Memory Only
4. All of These

Q6. This instruction format belongs to **1M**

31:25	24:20	19:15	14:12	11:7	6:0
func7	rs2	rs1	func3	rd	op

1. I-Type family
2. B-Type family
3. J-Type family
4. None of these

Q7. Contents of s1 after the execution of instruction `xori s1, zero, 0xFFFFFFFF` is **1M**

1. 0x1111 1111
2. 0xFFFF FFFF
3. 0x0000 0000
4. None of the above

Q8. In RISC-V vector assembly, if $VLEN=128$ bits, $LMUL=2$, $SEW=32$ bits, then the vectors at the application level can work with **1M**

1. 4 elements
2. 8 elements
3. 16 elements
4. None of these

Q9. In Figure 2, if the circled multiplexor in the WriteBack stage is pushed back into the Memory stage, then the clock period will **1M**

1. Increase

Figure 2

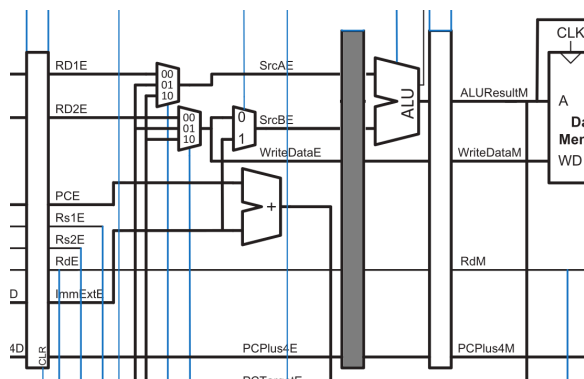
The diagram illustrates the MIPS processor architecture, showing the flow of data and control signals between various components. Key components include:

- Instruction Memory (A):** Receives `PCF` and `PCF` signals, outputs `InstrD` to the `Register File`.
- Register File:** Receives `InstrD` and `ImmSrcD1:6` signals, outputs `RD1`, `RD2`, `RD3`, and `WD3` signals to the `ALU` and `Data Memory`.
- ALU (B):** Receives `RD1`, `RD2`, `RD3`, and `WD3` signals, outputs `ALUResultM` to the `Data Memory`.
- Data Memory:** Receives `ALUResultM` and `WriteDataM` signals, outputs `RdM` and `RdW` signals to the `PCPlus4W` and `PCPlus4E` units.
- Control Unit:** Receives `RegWriteD`, `ResultSrcD1:3`, `MemWriteD`, `JumpD`, `BranchD`, `ALUControlD2:3`, `ALUSrcD`, and `ImmSrcD1:6` signals, outputs `RegWriteE`, `ResultSrcE1:3`, `MemWriteE`, `JumpE`, `BranchE`, `ALUControlE2:3`, and `ALUSrcE` signals to the `Register File` and `ALU`.
- PCPlus4F:** Receives `PCF` and `PCF` signals, outputs `PCPlus4F` to the `Register File`.
- PCPlus4D:** Receives `PCF` and `PCF` signals, outputs `PCPlus4D` to the `Register File`.
- PCPlus4E:** Receives `PCF` and `PCF` signals, outputs `PCPlus4E` to the `Register File`.
- PCPlus4W:** Receives `PCF` and `PCF` signals, outputs `PCPlus4W` to the `Register File`.

The diagram also shows the flow of data and control signals between various components, including the `Register File`, `ALU`, `Data Memory`, and various control units. The diagram is divided into several sections: Instruction Memory (A), Register File, ALU (B), Data Memory, and various control units. The diagram shows the flow of data and control signals between various components. Key components include: Instruction Memory (A), Register File, ALU (B), Data Memory, and various control units. The diagram shows the flow of data and control signals between various components. Key components include: Instruction Memory (A), Register File, ALU (B), Data Memory, and various control units.

3. Be unaffected
4. Be infinite

```
L7: lw x6, -4(x9)
    sw x6, 8(x9)
    or x4,x5,x6
    beq x4, x4, L7
```



Q10. If a new pipelined register is introduced as shown in Figure 3, then the clock period will be **1M**

1. Increase
2. **Decrease**
3. Be unaffected
4. Be infinite

- 1. 0**
2. 2
3. 4
4. None of the rest

1. Only A
2. Only B
- 3. Both A and B**
4. Both A and B, as well as ALU

Q13. If the $T_c = 0.1 \times (\text{sum of erp digits})$ nanoseconds and CPI=1 with 1 billion instructions, the program execution time will be **1M**

1. Greater than 1 second
2. Less than 1 second
3. Equal to 1 second

Q14. A RISC-V vector program executes 6 instructions with the following profile in Table 1. What is the average CPI? **1M**

1. 4.5
2. 5.0

3. 5.83

4. 6.25

Table 1

Instruction	Count	CPI
vsetvli	1	1
vle32.v	2	8
vfmacc.vv	1	6
vfadd.vv	1	4
vse32.v	1	8

Q15. Assume clock frequency = 2 GHz.
According to Table 1, what will be the total program execution time? **1M**

- 10.5 nanoseconds
- 14 nanoseconds
- 16 nanoseconds
- 17.5 nanoseconds**

Q16. If the hazard unit has a delay of $t_{pd}=10 \times$ (sum of student erp digits) ps, will the pipeline work correctly at $T_c=520$ ps? **1M**

Element	Parameter	Delay (ps)
Register clk-to-Q	t_{pcq}	40
Register setup	t_{setup}	50
Multiplexer	t_{mux}	30
AND-OR gate	t_{AND-OR}	20
ALU	t_{ALU}	120
Decoder (control unit)	t_{dec}	25
Extend unit	t_{ext}	35
Memory read	t_{mem}	200
Register file read	t_{Rfread}	100
Register file setup	$t_{Rfsetup}$	60

- Yes
- No
- Not enough information

Q17. In Figure 4, if $VLEN=256$ bits, $AVL=6$, then determine how many iterations will be used to accomplish the task in code **1M**

- 1**
- 2
- 3
- 4

Q18. In figure 4, if $VLEN=64$ bits, $AVL=6$, then will the `slli t0, t0, 2` be replaced with

- `slli t0, t0, 1`
- `slli t0,t0, 3`
- `slli t0, t0, 4`
- None of the rest**

```
x: .word 10,20,30,40,50,60
y: .word 5,15,25,35,45,55
z: .word 0, 0, 0, 0, 0, 0

vvaddint32
vsetvli t0, a0, e32,
    m1, ta, ma
vle32.v v0, (a1)
sub a0, a0, t0
slli t0, t0, 2
add a1, a1, t0
vle32.v v1, (a2)
add a2, a2, t0
vadd.vv v2, v0, v1
vse32.v v2, (a3)
add a3, a3, t0
bnez a0, vvaddint32
ret
```

Figure 4

Q19. Consider a 5-stage RISC-V pipeline (IF, ID, EX, MEM, WB) with forwarding and hazard detection units implemented. There is no branch prediction. Assume the following instruction sequence:

```
lw x2, 0(x1)
mul x3, x2, x4
add x5, x3, x2
beq x3, x0, label #taken
and x6, x5, x3
```

Which of the following is the minimum number of stall cycles required for the correct execution of this instruction sequence, and why? **1M**

- 1 stall cycle, due to a single load-use hazard.**
- 2 stall cycles, due to the load-use hazard and the branch dependency.
- 3 stall cycles, accounting for load-use hazard, branch resolution, and forwarding delays.
- 4 stall cycles, considering both load-use and branch hazards along with data forwarding limitations.

Q20. The Throughput of a processor is **1M**

- Cycles per instruction
- Instructions per cycle**
- Cycles per two instructions
- Instructions per two cycles

Q21. The following instruction is located at address 0x00400020

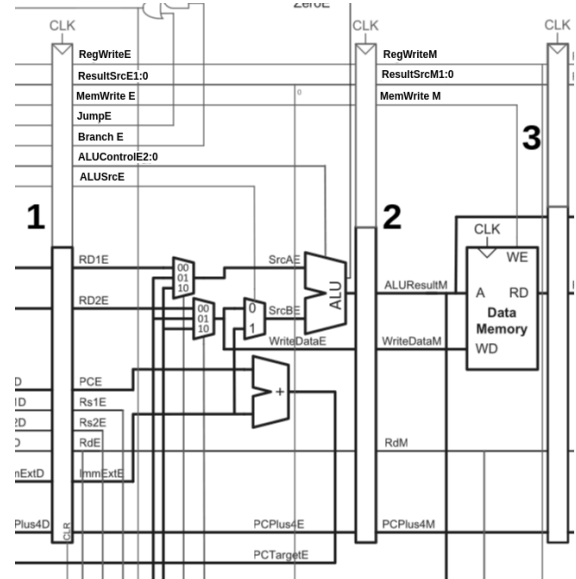
beq x1, x2, -16

What is the Branch Target Address? **1M**

1. 0x00400000
2. **0x00400010**
3. 0x00400030
4. 0x00400034
- Q22. The following instruction is located at address 0x00400100
jal x0, 64
What is the Jump Target Address? **1M**
1. 0x00400040
2. **0x00400140**
3. 0x00400240
4. None of the rest
- Q23. In a hypothetical case, if there is an assumption of two hidden bits in branch size accommodation in the immediate field, then which of the following correctly defines the output of the extend unit **1M**
1. $PC + \text{SignExt}(\{\text{imm}_{12:2}, 1'b0\})$
2. $PC + \text{SignExt}(\{\text{imm}_{12:1}, 1'b0\})$
3. $PC + \text{SignExt}(\{\text{imm}_{12:1}, 2'b00\})$
4. **$PC + \text{SignExt}(\{\text{imm}_{13:2}, 2'b00\})$**
- Q24. The following instruction is at address 0x00400080:
auipc x5, 2
whose working principle is
auipc rd, upimm: $rd = \{\text{upimm}, 12'b0\} + PC$
What will be the value stored in x5? **1M**
1. **x5 = 0x00402080**
2. x5 = 0x00402000
3. x5 = 0x00400080
4. None of the rest
- Q25. Highlight the data and control signals' path in Figure 2 for **auipc x5, 2**. Note: You will get **1M** only if you correctly highlight 4 paths
1. Yes (For teacher use only)
2. No (For teacher use only)
- Q26. What is the range of go-to addresses in lw and sw instructions in RISC-V in terms of instructions? **1M**
1. -2048 to 2047
2. **-512 to 511**
3. -4096 to 4095
4. None of the rest
- Q27. Architecture and Microarchitecture are **1M**
1. Hardware and Instruction Set
2. Instruction Set and Software
3. Software and Hardware
4. **Instruction Set and Hardware**

Q28. Which pipeline register, in Figure 5, has the maximum number of flip-flops? **1M**

1. **1**
2. 2
3. 3
4. All have an equal number of flip-flops



Q29. The register file _____ in single cycle and _____ in pipelined implementation **1M**

1. Negative edge triggered, positive edge triggered
2. **None of the rest**
3. Positive edge triggered, positive edge triggered
4. Negative edge triggered, negative edge triggered

Q30. A program has 20% branch instructions. The scalar pipelined processor has a misprediction rate of 10% and a penalty of 2 cycles. Ideal CPI = 1. What is the effective CPI? **1M**

1. 1.01
2. **1.04**
3. 1.20
4. 1.40

Q31. A 2x2 matrix is stored in row-major order in memory starting at base address x1 = 0x2000

Original Matrix A: Memory Layout:

1	2
3	4

Address 0x2000 = 1 (A[0][0])
Address 0x2004 = 2 (A[0][1])
Address 0x2008 = 3 (A[1][0])

Address 0x200C = 4 (A[1][1])

To transpose this matrix, the following code is executed:

x1 = 0x2000 (base address)

Each element is 4 bytes (32-bit integer)

```
vsetivli x0, 4, e32, m1
vluxei32.v v2, (x1), v1
```

Which index vector v1 (in byte offsets) correctly gathers elements to produce the transposed matrix {1, 3, 2, 4} in v2? **1M**

1. v1 = {0, 4, 8, 12}
- 2. v1 = {0, 8, 4, 12}**
3. v1 = {4, 0, 12, 8}
4. v1 = {12, 8, 4, 0}

Q32. In jal instruction, the jump address is _____, and the link address is _____ **1M**

1. PC+4, JTA
- 2. JTA, PC+4**
3. JTA/2, PC+4
4. None of the rest

Q33. Program execution time in pipelined implementation reduces due to **1M**

1. Increased T_c , Decreased CPI
- 2. Increased CPI, Decreased T_c**
3. Increased CPI, Increased T_c
4. Decreased CPI, Decreased T_c

Q34. There are 4 operational stages in RISC-V pipelined implementation **1M**

1. True
- 2. False**

Q35. The following instructions respectively belong to **1M**

lw x6, -4(x9)

sw x6, 8(x9)

or x4, x5, x6

beq x4, x4, L7

1. S, I, B, R
- 2. I, S, R, B**
3. I, S, B, R
4. S, I, R, B

Q36. What family of instructions do these hex codes represent? **1M**

0064A423

FFC4A303

FE420AE3

0062E233

- 1. S, I, B, R**
2. I, S, B, R
3. S, I, R, B
4. I, S, R, B

Q37. Which operational stage does the control signal RegWriteM belong to **1M**

1. Writeback stage
2. Execute stage
3. Decode Stage
- 4. None of the rest**

Q38. For the code

```
addi s0, zero, 6
addi s1, zero, 0
```

LOOP:

```
beq s0, zero, DONE
lw s2, 0(s1)
lw s3, 16(s1)
lw s4, 32(s1)
lw s5, 48(s1)
addi s0, s0, -1
j LOOP
```

DONE:

What will be stored in the immediate field of the beq instruction? **1M**

1. 28
2. 7
- 3. 14**
4. None of the rest

Q39. RISC-V is **1M**

1. IP of Intel
- 2. Open-source**
3. IP of AMD
4. None of the rest

Q40. RV32I is **1M**

1. Vector Extension
- 2. Base ISA**
3. Integer Extension
4. None of the rest

*****The End*****